



AI Fellowship For Global Young Innovators 2026

AUTONOMOUS AI-DRIVEN HUMAN DETECTION IN HIGH-RISE BUILDING FIRES USING YOLO

P R E S E N T A T I O N

Domain:
Identification of Humans in Hazardous
Environments

By:
Kishan Kannan

Email:
kishankannan.blr@gmail.com



A place where life can't survive
This is not deep space. It's a fire scene

Firefighters run towards the
flames when everyone else is
running out.

The Problem I'm Solving For

01

Sub Problem 1

High-rise fires compromise the integrity of load-bearing structures, resulting in unstable conditions for firefighters.

02

Sub Problem 2

Exposure to carbon monoxide and superheated gases can cause firefighters to experience rapid incapacitation as well as long-term health consequences.

03

Sub Problem 3

Dense smoke and buildings debris act as a wall between firefighters and victims who need to be rescued.

Current Solutions In Market

01

Solution 1

One key technology speeding up response is digital alerts, which bypass the 911 call.

Once a monitoring station detects a fire, it automatically alerts the closest and most well-equipped units

02

Solution 2

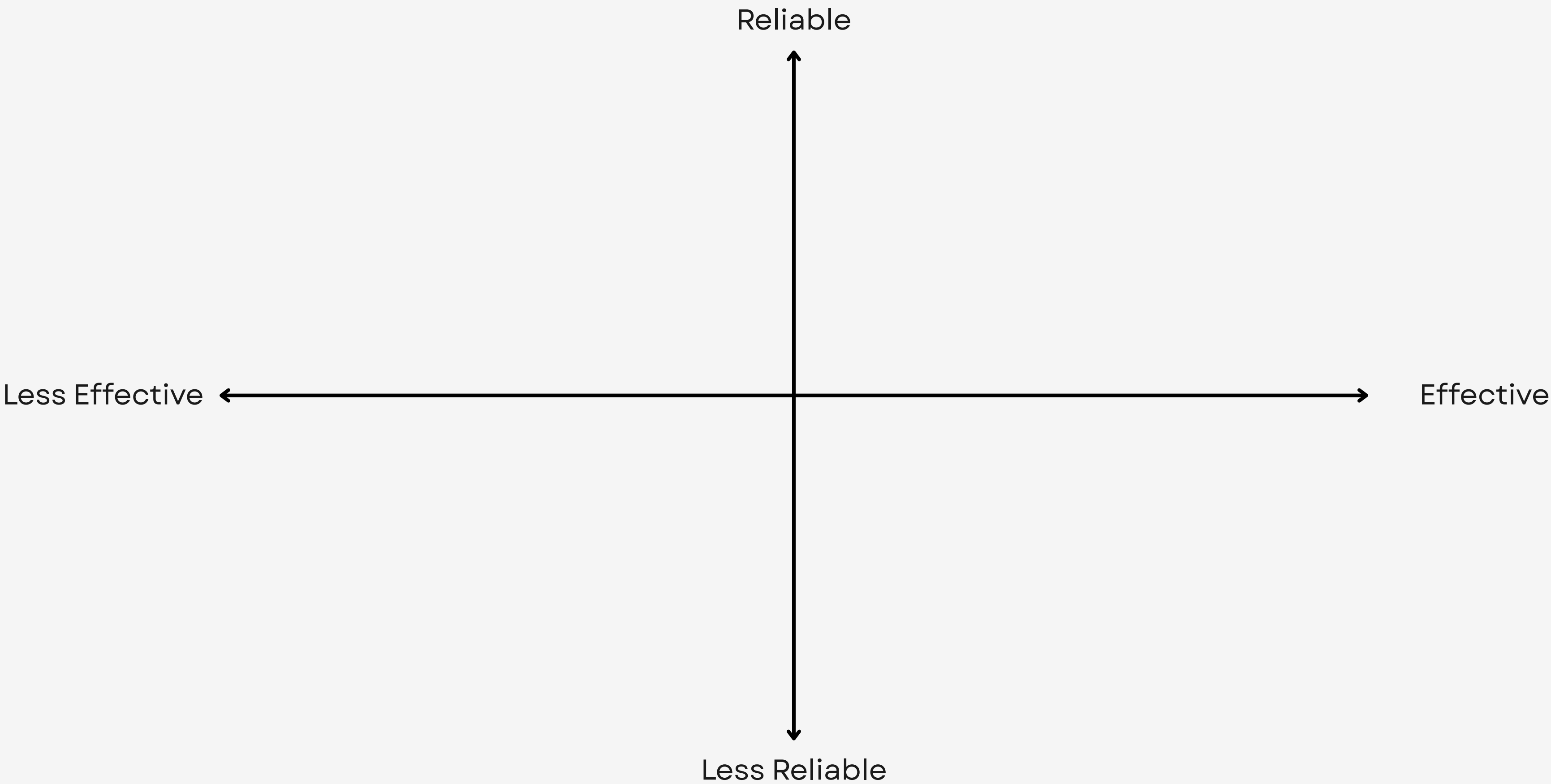
Graduate innovation of more sophisticated fire protective fabrics which are used in the firefighter apparel for heat protective outer garments as well as flame protective fabrics

03

Solution 3

The synthesis of advanced fire-resistant foam formulations designed for combating flammable liquid fires represents an effective firefighting system.

Competitive Landscape



My Solution

01

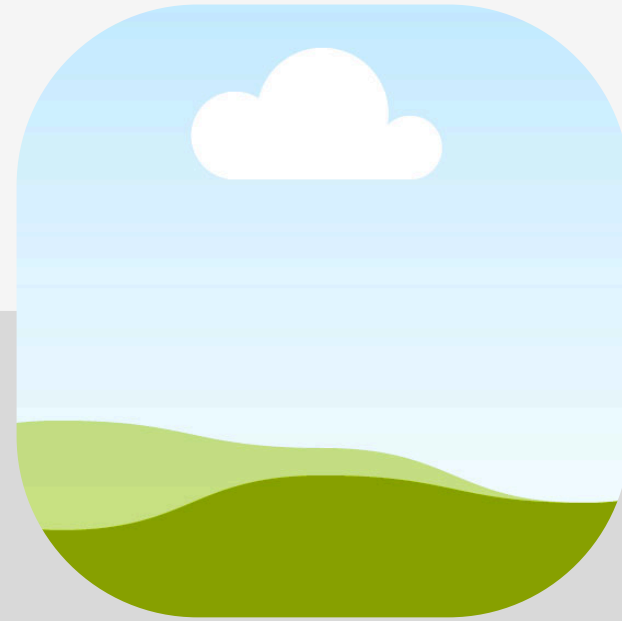
- The ability to see in visible light is severely compromised when a smoke column turns black. However, a thermal-tuned YOLOv8 model, operating on the drone's edge GPU, consistently identifies individuals frame by frame, maintaining confidence scores sufficient for actionable responses.
- Transmitting raw 4K and thermal video to a ground station and processing it with YOLO would incur a round-trip delay of 3 to 5 seconds over a degraded radio link. In that time, a trapped individual could potentially move 20 meters. Therefore, the model operates directly on the drone.

Target Audience



User 1

- Who will benefit from your solution?
- Age group, region, demographics, etc.



User 2

- Who will benefit from your solution?
- Age group, region, demographics, etc.



User 3

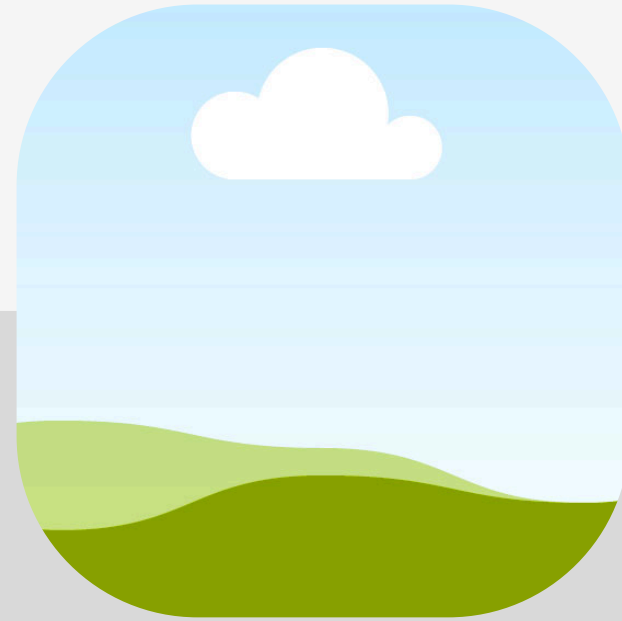
- Who will benefit from your solution?
- Age group, region, demographics, etc.

Stakeholders Involved



User 1

- Who will be involved or affected? (e.g., users, investors, suppliers, institutions)
- Mention any partnerships or ecosystem players



User 2

- Who will be involved or affected? (e.g., users, investors, suppliers, institutions)
- Mention any partnerships or ecosystem players



User 3

- Who will be involved or affected? (e.g., users, investors, suppliers, institutions)
- Mention any partnerships or ecosystem players

My Big Vision

Vision

To build an autonomous AI model for rescue bots and drones, enabling them to intelligently identify humans in hazardous fire conditions within high-rise buildings. This innovation aims to enhance the safety and efficiency of rescue operations, minimizing risk to human responders.

Mission

To create a python based YOLO_v8 or YOLO family model to identify humans in harardous environments like a fire scene, under critical conditions like thick smoke or dense buildings debris.



Key Features

Feature 1

Visualize your competitive advantages using a quadrant for easy scanning.

Feature 2

How is the landscape differentiated and what makes you different.

Feature 3

How is the landscape differentiated and what makes you different.

Feature 4

What can you do better than your competition?
How will you outperform them?

My Big Differentiator

01

Differentiator 1

Give a striking overview of the problem and explain it briefly.

02

Differentiator 2

Elaborate on how this negatively impacts people and their experiences.

03

Differentiator 3

Frame the solution effectively as it will set the stage of your entire pitch.

Go-To-Market Strategy

Customer Acquisition
Channels

Acquisition Strategy

Sales Model

Revenue Model

How will you earn money? (Revenue model)

What are your major costs? (Fixed and variable expenses)

Scalability Plan

01

Plan 1

Describe how you envision to solve the problems you previously shared.

02

Plan 2

Describe how you envision to solve the problems you previously shared.

03

Plan 3

Describe how you envision to solve the problems you previously shared.

Graph or diagram

5 Year Plan

- Project your venture’s journey over the next five years
- Milestones, growth stages, and long-term goals

Graph or diagram

Impact Assessment

What social, economic, or environmental impact will your solution create?

How will you measure success?

Chart or Graph



AI Fellowship For Global Young Innovators 2026

Thank You
FOR YOUR TIME

Domain

Your Name

Your Email